SASHA
Skies and Seas Hydrogen-fuels Accelerator Coalition

Consultation response

Response to the Draft Icelandic Employment Policy – Growth Plan to 2035 | December 2025

E-kerosene as an indisputable economic opportunity

With its large capacity for renewable electricity, Iceland is in an excellent position to bring market competitive e-fuels to Europe. Europe is poised to need large amounts of e-fuels as its objectives for 2030 and 2035 come into view. Seeking to become more sovereign on energy matters, the EU will favour production in the European Economic Area (EEA). Within Europe, Iceland is one of just a handful of countries that can generate competitively priced renewable hydrogen, and therefore the e-fuels that Europe needs. Iceland should use this opportunity to ramp up its production capacity, choosing squarely for renewable hydrogen and e-fuels now, and becoming a key European hub for decarbonisation.

The regulatory context

To correctly address the elements likely to generate growth for the Icelandic economy, it is important to consider what one of its key markets is planning for the near future. In the European Union (EU), where aviation and shipping are major economic players, mandates have been set for the use of alternative fuels to decarbonise the sectors in the coming decades.

The legislations that specify this are a mix of the European Emissions Trading System or EU ETS[1], FuelEU Maritime (FEUM)[2], and ReFuelEU Aviation (RFEUA)[3]. The former is a cap-and-trade system with decreasing amounts of greenhouse gas emission allowances, which incentivises the sector to gradually decarbonise, while the latter two set several obligations around a fuel switch for aviation and maritime

[1] https://eur-lex.europa.eu/eli/dir/2003/87/oj/eng
[2] https://eur-lex.europa.eu/eli/reg/2023/1805/oj/eng
[3] https://eur-lex.europa.eu/eli/reg/2023/2405/oj/eng

with a specific mandate for renewable fuels of non-biological origin (RFNBO, or e-fuels).

Two key aims are central here: improving the sustainability of the sectors and guaranteeing the energy sovereignty of the continent.

Further pushing on these two key aims is the recently published Sustainable Transport Investment Plan (STIP)[4] which marked a significant step in sending a positive investment signal and helping to kick-off the market: 2,9 billion EUR will be pooled between 2025-2027 to support renewable fuels in the aviation and maritime sectors. To answer access to market challenges, the STIP also confirmed the launch of a double-sided auction pilot for e-kerosene, which is set to pave the way for a future EU-wide mechanism aimed also at maritime fuels.

With the current feedstock for fossil fuels being almost entirely imported from outside of Europe[5], Europe is very vulnerable to geopolitical volatility. Aware of this, the EU is therefore actively working to ramp up alternatives to fossil kerosene and shipping fuels and has taken a decisive step in that direction with its RFEUA and FEUM targets and the STIP for so-called sustainable aviation fuels (SAF) and sustainable maritime fuels (SMF).

What are SAF & SMF

Alternative fuels, often referred to as SAF and SMF, are not made equal. Simply put, there are two main types of alternative fuels: biofuels, produced using biomass, and e-fuels, made from electrolytic hydrogen and a source of carbon. The fuels emit the same amount of carbon as fossil fuels at combustion, so their ability to lower emissions is entirely based on the full lifecycle emissions of the fuel and the feedstock used to produce it.

For biofuels, first generation crop-based biofuels (produced using crops grown specifically to be used as a fuel) and used cooking oils and animal fats are available in such limited quantities that they don't present a long-term, viable route to producing alternative fuels[6]. Crop-based fuels also come with additional land use impacts[7], while used oils and fats have competing, existing uses. Moreover, biofuels currently only present 20-40% emission reductions[8]. Contrary to biofuels, whose limited feedstock will generate a massive price increase as soon as limits start coming into view, e-fuels costs will be trending downwards over time, as the effects

[4] https://transport.ec.europa.eu/document/download/73447373-de2a-4ba4-9371-36d1186035d4_en?filename=COM_2025_664_STIP.pdf
[5] https://ec.europa.eu/eurostat/statistics-explained/SEPDF/cache/43212.pdf
[6] https://www.argusmedia.com/en/news-and-insights/latest-market-news/2493411-biofuel-production-limited-by-feedstocks-in-2030-iea
[7] https://www.sashacoalition.org/biodiversity-risks-eu-aviation-maritime-policy
[8] https://www.transportenvironment.org/articles/biofuels-in-cars-a-dead-end-for-europe

of scaling set in[9]. Moreover, biofuels currently are mostly import dependent[10] and are unlikely to offer the needed change to reassure the EU's aspirations to become more energy independent.

E-fuels, however, must be produced using green hydrogen and a sustainable source of carbon dioxide[11] like Direct Air Capture under the EU Delegated Act defining RFNBOs[12]. If a renewable source of electricity is used, this makes them close to carbon neutral. However, producing e-fuels in the quantities needed requires significant amounts of renewable electricity.

The opportunity for Iceland

This is where Iceland's role is crucial in the build out of this market. Iceland's sizable capacity for renewable electricity allows it to bring market competitive e-fuels to Europe. Iceland is one of just a handful of countries that have reached the low-carbon grid threshold of 18 gCO2eq/MJ. This allows projects to even develop e-fuels on the grid that comply with the RFNBO mandates[13]. With the European focus squarely on the EEA in terms of the financial support it plans to provide (outlined in the European Commission's STIP), imports from outside the EEA will not be favoured and EEA-based production will be able to count on significant regulatory and financial support.

While there are questions around the economic benefits of unbridled aviation growth[14], the sector is set to continue growing for the time being[15]. The same is expected of the maritime sector[16]. Even without such growth, the amounts of e-fuels needed in the years to come will be a boon for the Icelandic economy, should it choose to welcome this opportunity.

In 2030, a forecasted 2.2 Mt of kerosene will legally have to be e-kerosene (or 1,2% of the total share of kerosene). That volume climbs to over 16Mt in 2035 (or 5% of total share). The mandate for e-kerosene goes up to 35% of the total volume in 2050[17].

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[9] https://www.fathom-consulting.com/highlights-from-the-report/#:~:text=Fathom's%20analysis%20indicates%20a%20far,economy%2C%20geopolitics%20and%20financial%20markets.
[10] https://www.easa.europa.eu/en/newsroom-and-events/press-releases/easa-publishes-report-sustainable-aviation-fuel-scale-progress
[11] With the exception of e-ammonia that requires nitrogen feedstock instead of carbon
[12] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32023R1185
[13] https://www.transportenvironment.org/uploads/files/202504_e-kerosene_report.pdf
[14] https://www.transportenvironment.org/articles/new-analysis-debunks-the-assumption-that-air-passenger-growth-drives-economic-growth
[15] https://www.eurocontrol.int/sites/default/files/2024-12/eurocontrol-aviation-long-term-outlook-flights-co2-forecast-2024-2050.pdf
[16] https://ecsa.eu/shipping-industry-calls-for-action-following-the-attack-on-the-mv-tutor-and-cargo-ship-verbena-2-2/#:~:text=Published%20On:%20September%209%2C%202024,under%20the%20ETS%20Innovation%20Fund.
[17] https://transport.ec.europa.eu/transport-modes/air/environment/refueleu-aviation_en

Overall, for both shipping and aviation, an expected 6,8 Mt of e-fuels will be needed by 2035[18].

Staying ahead of the game

One of Iceland's key e-fuel project is SASHA member IðunnH2, based near Keflavík Airport, at the Helguvík Harbour. It is projected to produce 70.000 tons of e-kerosene per year[19]. Another Icelandic e-fuels project is Fjarðarorka and its planned 220.000 tons of e-ammonia[20]. Both these projects leave ample room for scale-up on a market that is clearly defined already in Europe and will likely gain in traction in the rest of the world as more regions implement sustainability obligations for the aviation sector.

A clear endorsement of e-fuels by the Icelandic government, in line with science and the European regulatory context, would be a local market signal that creates more certainty for investors; an aspect many e-fuel producers throughout the European region are struggling with, despite the unambiguous policy route Europe has taken.

By making its choice now for e-fuels and renewable hydrogen over biofuels, Iceland will take a competitive lead by opting for the energy carrier of the future, over one that is 1) limited in quantity and therefore set to quickly increase in price, 2) does not provide sufficient decarbonisation advantages and comes with serious biodiversity and fraud risks, and 3) does not provide the sovereignty benefits that e-fuels do. By adding an early start to the renewable price advantage, Iceland can take full advantage of a double lead over other European countries working to build up e-fuels production capacity.

[18] https://transport.ec.europa.eu/news-events/news/commission-unveils-sustainable-transport-investment-plan-strategic-approach-boost-renewable-and-low-2025-11-05_en#:~:text=The%20Investment%20Plan%20responds%20to,by%202035%20to%20drive%20production.
[19] https://idunnh2.com/methanol-to-jet-viability-confirmed-for-icelands-largest-e-saf-project/#:~:text=Carbon%20Recycling%20International%20(CRI)%20has%20completed%20a,produce%20approximately%2070%2C000%20tons%20of%20e%2DSAF%20annually.
[20] https://fjardarorka.is/en/

About the SASHA Coalition

The Skies and Seas Hydrogen-fuels Accelerator (SASHA) Coalition unites companies across the shipping and aviation value chains to advocate for policy supporting renewable hydrogen climate solutions.

We work collaboratively with industry and policymakers in Europe to help develop and commercialise the most sustainable technologies for these hard-to-abate sectors at the pace the climate crisis requires.

The coalition is facilitated by climate NGO Opportunity Green.

For more information, please contact:

Aurelia Leeuw
Director of EU Policy
Opportunity Green & SASHA Coalition
aurelia@opportunitygreen.org